Urologic Sepsis

Section 1: Case Summary

Scenario Title:	Urologic Sepsis and Shock
Keywords:	Urologic emergency, sepsis, septic shock
Brief Description of Case:	Denise is a 59-year-old female who presents with a 7-day history of urinary symptoms, fever, and started with left flank pain. She has a history of STEMI 5 years ago. She then becomes unstable requiring fluid resuscitation, vasopressors, and empiric antibiotic treatment. The team leader will manage a patient with severe sepsis secondary to an infected ureteric calculus then arrange emergent urologic consultation and admission to hospital.

Goals and Objectives	
Educational Goal:	Review the initial assessment and management of a patient with acute flank pain and fever.
Objectives: (Medical and CRM)	1. Early recognition of sepsis and identifying a likely source of infection 2. Timely implementation of broad-spectrum antibiotics and source control 3. Eliminate other differential diagnoses of shock 4. Management of a hemodynamically unstable patient with fluid resuscitation, vasopressors and appropriate monitoring 5. Appropriate hospital disposition of the patient
EPAs Assessed:	

Learners, Setting and Personnel				
Target Learners:	☒ Junior Learners	☐ Senior Learners	☐ Staff	
	☒ Physicians	☐ Nurses	☐ RTs	☐ Inter-professional
	☐ Other Learners:			
Location:	☒ Sim Lab	☐ In Situ	☐ Other:	
Recommended Number of Facilitators:	Instructors: 1			
	Confederates: 1			
	Sim Techs: 1			

Scenario Development	
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Last Revision Date:	July 27, 2020
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Version Number:	5.0

Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Denise Koffman			Age: 59	Gender: F	Weight: 90kg
Presenting complaint: fever and urinary symptoms					
Temp: 39.0	HR: 95	BP: 100/65	RR: 24	O_2Sat: 96%	FiO_2: 21%
Cap glucose: 9.6			GCS: 15		
Triage note: 59-year-old female with 7-day history of dysuria, urinary frequency, and fever. This was preceded by intermittent left flank pain that has now become constant. She called EMS due to worsening left flank pain, nausea, and decreased PO intake.					
Allergies: NKDA					
Past Medical History: STEMI (5 years ago) Hypertension Hyperlipidemia			Current Medications: Ramipril 4mg PO daily Atorvastatin 10mg PO daily Metoprolol 50mg PO BID Lasix 20mg PO daily Aspirin 81mg PO daily		

Section 2B: Extra Patient Information

A. Further History
Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, SP, etc.)? - Patient emphasizes that intermittent flank pain preceded fever and dysuria - Patient had an inferior STEMI and stent 5 years ago. She has been followed by her community cardiologist and her most recent echocardiography showed LVEF >45% (4 years ago). - No urologic history, review of systems is otherwise unremarkable. - Patient lives at home with husband. Retired teacher. Remote smoking (<10 pack-year), social EtOH, and no recreational drug use

B. Physical Exam	
List any pertinent positive and negative findings	
Cardio: borderline tachycardia, normal heart sounds	Neuro: appears anxious and unwell, no focal neurologic deficits
Resp: GAEB, no wheezing or crackle	Head & Neck: unremarkable
Abdo: Soft, mild left tenderness without guarding	MSK/skin: Cap refill 4-5 sec. No skin changes/bruise
Other: Significant left CVA tenderness	

Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin: adult
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Standard emergency department resuscitation supplies Bedside ultrasound machine
C. Required Medications
Ringer’s lactate or normal saline Broad spectrum antibiotics (e.g. Piperacillin-Tazobactam or ceftriaxone/ampicillin/gentamycin) Norepinephrine (or equivalent vasopressors) Analgesics Anti-emetics
D. Moulage
None
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed
☒ Patient not yet on monitor
F. Patient Reactions and Exam
Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format. Airway: patent, speaking in full sentences Breathing: RR 24 no active respiratory distress, O2 sat normal Circulation: borderline tachycardia, delayed capillary refill time Disability: GCS 15, appears anxious and unwell Exposure: Moaning when L flank palpated or CVA angle manipulated.

Section 4: Confederates and Standardized Patients

Sim Actors and Standardized Patient Roles and Scripts	
Role	*Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)*
Bedside nurse	Calls team leader to bedside because the patient looks unwell. RN is skilled and helpful.

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: sinus rhythm HR: 95 BP: 100/65 RR: 20 O_2SAT: 96% T: 39^{o}C GCS: 15	*Appears unwell, delayed capillary refill*	<u>Expected Learner Actions</u> ☐ Recognize patient likely has urosepsis ☐ Making sure patient is protecting airway and breathing ☐ 1-2L bolus of fluid, then reassess volume status ☐ Bloodwork, including urine and blood cultures ☐ Broad spectrum antibiotics ☐ Insert urinary catheter	<u>Modifiers</u> *Changes to patient condition based on learner action* - EKG/CXR provided <u>Triggers</u> *For progression to next state* -Once the team identifies sepsis, starts IV fluids/continuous monitoring, and starts antibiotics OR after 5 minutes	e.g. pip-tazo 3.375-4.5g IV OR ceftriaxone 1-2g IV OR ampicillin+ gentamycin
2. Rhythm: sinus tachy HR: 120 BP: 80/55 RR: 22 O_2SAT: 95% T: 39^{o}C GCS: 14	*Appears drowsy and just moaning*	<u>Expected Learner Actions</u> ☐ Reassess ABCs and volume status and recognize patient likely has septic shock ☐ Consider doing bedside ultrasound (Cardiac, IVC, Hydronephrosis) ☐ Consider more IV fluids vs direct to vasopressors ☐ Start vasopressor ☐ Central line or second large IV ☐ Insert urinary catheter ☐ Order CT abdo/pelvis or ultrasound once patient is stabilized	<u>Modifiers</u> - Patient declines if no second fluid bolus/vasopressor started (increase HR to 125, BP to 70/40) - - <u>Triggers</u> - Once the team identifies shock, initiates more fluids, and vasopressors OR after 5 minutes -	The trainee should be prompted if appropriate treatment has not been initiated. Consider norepinephrine 0.1-2mcg/kg/min IV as first line

3. Rhythm: sinus rhythm HR: 85 BP: 110/75 RR: 18 O_2SAT: 95% T: 37.8°C GCS: 15	Improved mental status, normalized capillary refill, and GCS 15	Expected Learner Actions ☐ Interpret CT results ☐ Consider doing bedside renal ultrasound (show same image as the above pointer) ☐ Consult urology and ICU	Modifiers Triggers **END CASE** after appropriate management of septic shock (abx, fluids, vasopressors) and disposition	

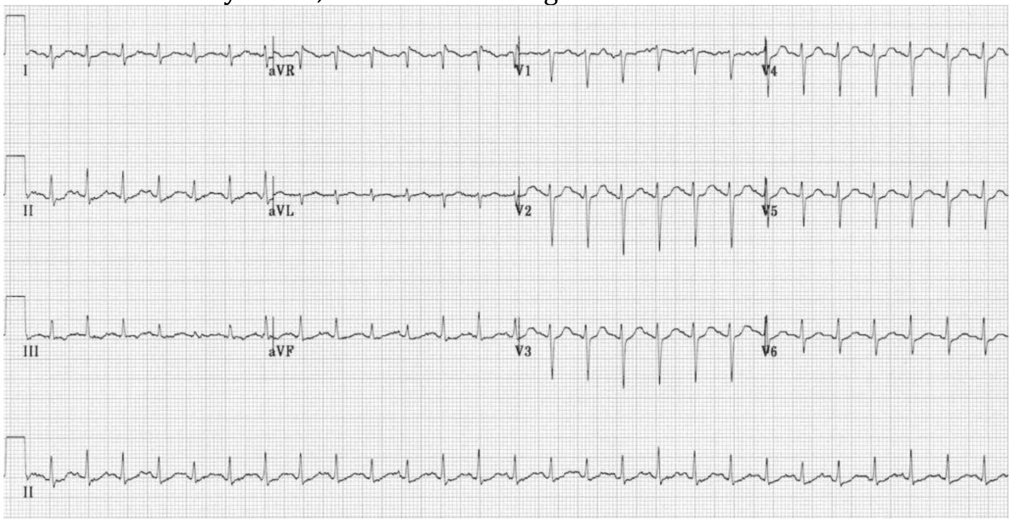
Appendix A: Laboratory Results

CBC WBC 20.3 Hgb 137 Plt 600 Lytes Na 135 K 5.1 Cl 98 HCO_3 16 AG 21 Urea 15.1 Cr 217 Glucose 9 VBG pH 7.26 pCO_2 32 pO_2 55 HCO_3 16 Lactate 3.8	Cardiac/Coags Trop <12 INR 1.1 Biliary AST 49 ALT 33 GGT 60 ALP 105 Bili 8 Lipase 82 Other B-HCG negative Urinalysis Color: Cloudy Spec gravity: 1.030 RBC: 3+ WBC 2+ Nitrites: Positive

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

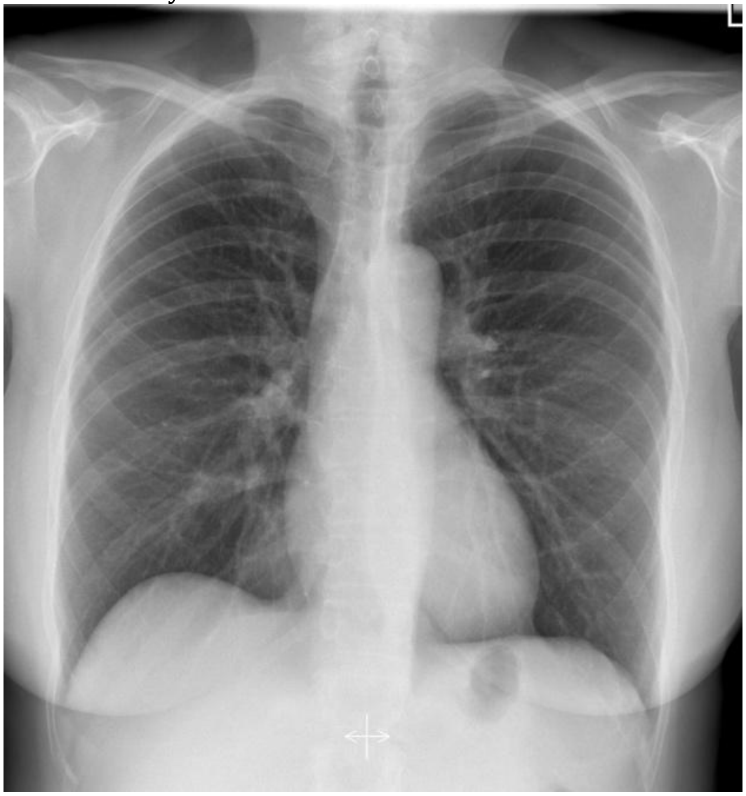
Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!

1. ECG: sinus tachycardia, no ischemic changes



From Life in the Fast Lane (https://litfl.com/sinus-tachycardia-ecg-library/)

2. Chest X-Ray: normal



Case courtesy of Dr Usman Bashir, Radiopaedia.org, rID: 18394

3. CT abdo/pelvis: Large (11 mm) obstructing calculus in the left ureter with left hydro-ureteronephrosis.

Image courtesy of Dr Roberto Schubert, Radiopaedia.org, rID: 16407

4. Ultrasound: Hydronephrosis

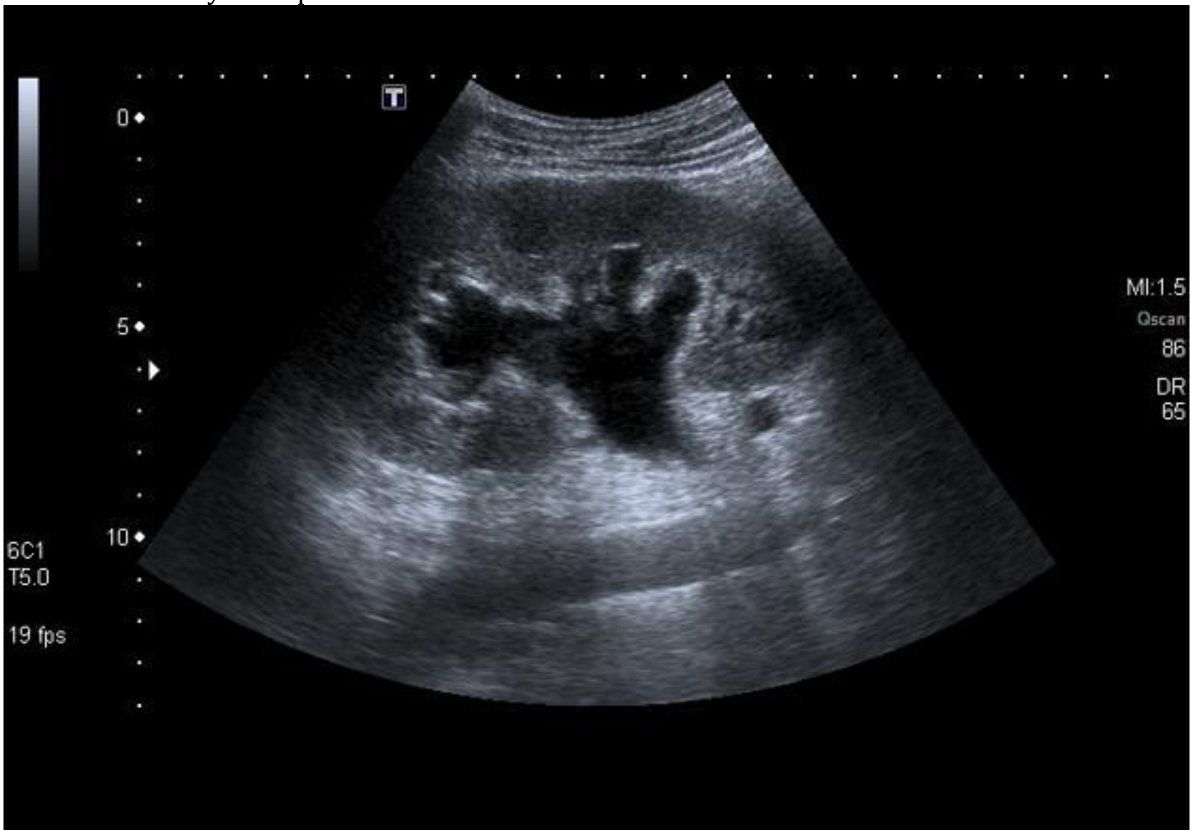


Image courtesy of Dr Amr Refat, Radiopaedia.org, rID: 25562

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

Discussion points for residents/trainees

- Can you summarize the case for us?
- How did the case go as a team? How did it go for you as an individual?
- What went well during the simulated resuscitation?
- What did not go so well? Anything that you would have done differently?

Medical Knowledge

- What is your approach to shock and how does POCUS help confirming your diagnosis?
    - Septic/distributive shock
    - Hemorrhagic/hypovolemic shock
    - Obstructive shock e.g. cardiac tamponade
    - Cardiogenic shock
    - Anaphylactic shock
    - POCUS
        - Cardiac tamponade
        - LV contractility (hypokinetic vs hyperkinetic)
        - RV dilation/strain and RV outflow tract obstruction
        - IVC compressibility and variation with respiration
        - FAST for free fluid
- What are your choices of vasopressors in response to septic shock? What are your considerations in the context of patient's cardiac history? Discuss vasopressor mechanism of actions.
    - Norepinephrine (good 1st choice for most patients with septic shock, mostly alpha, some beta-1)
    - Phenylepherine (pure alpha agonist; good for vasodilatory or distributive shock; similar outcomes in septic patients compared to norepi)
    - Epinephrine (powerful beta receptor agonist/inotropic agent; caution if systolic heart failure)
    - Vasopressin (pure vasoconstrictor, non-catecholamine; considered as an adjunct vasopressor for shock and may improve renal perfusion)
- What are your target goals of resuscitation (things to consider)?
    - Fluids and vasopressors to keep MAP >=65 while end-organ perfusion maintained
        - Consider POCUS, pulse pressure variation, etc. to guide management
    - Lactate clearance
    - Urine output vs capillary refill time normalization
- Were you concerned about the patient's initial HR being 95?
    - Yes, because patient was also taking beta-blocker, masking tachycardia
- How did the patient's cardiac history change your approach to fluids and vasopressors?
    - Careful fluid bolus, expect septic cardiomyopathy, early vasopressors.
- How would your team-based approach change during the COVID pandemic?
    - PPE precaution and consider COVID as a differential diagnosis

References

1. Tintinalli's Emergency Medicine. Chapters 50, 56, and 89
2. EM Cases. "Episode 122 Sepsis & Septic Shock – What Matters Live from EM Cases Course", with Dr. Sara Gray
3. Kane SP and Benken ST. "Vasopressor Selection in Septic Shock", Society of Critical Care Medicine
4. Farkas J. "Vasopressors", The Internal Book of Critical Care